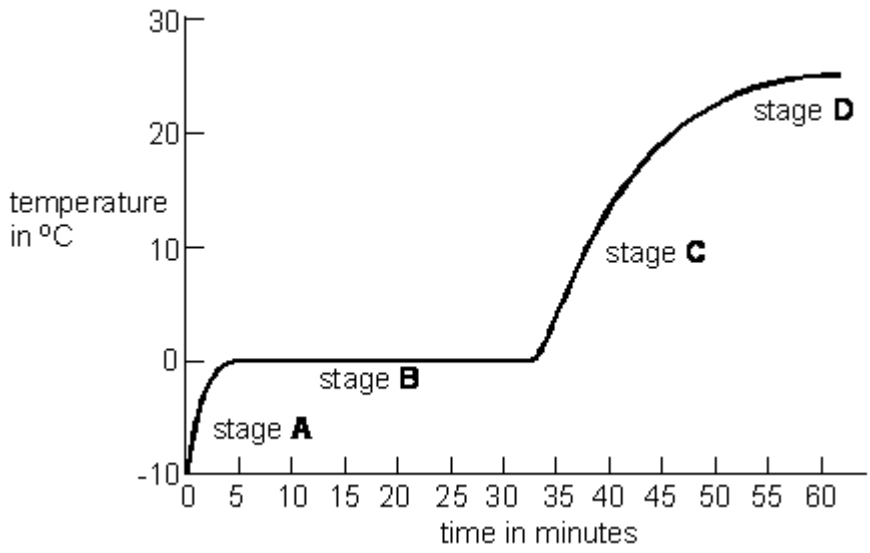
Q1.

A test tube of crushed ice is taken out of a freezer and left in a warm room. The graph shows how the temperature in the test tube changes.



(a) What is happening to the ice at stage **B**?

.....

1 mark

(b) Why does the temperature of the water stop rising at 23°C (stage **D**)?

.....

.....

1 mark

(c) Four descriptions of the ways molecules could move are given below.

They vibrate around fixed points. ☐

They move past each other and are close together. ☐

They move in straight lines, colliding occasionally. ☐

They all move in the same direction at the same speed. ☐

(i) How do the molecules move at stage **A**?
Write **A** in the correct box above.

1 mark

(ii) How do the molecules move at stage **C**?
Write **C** in the correct box above.

1 mark

(d) Ice from a freezer is put in a glass of water at room temperature. The ice floats in the water.

(i) What does this show about the density of the ice compared to that of water?

Tick the correct box.

Ice is more dense than water. ☐

Ice and water have the same density. ☐

Ice is less dense than water. ☐

Ice has a density of zero. ☐

1 mark

(ii) The fact that ice floats in water tells us something about the distances between the molecules.

Tick the box by the correct statement.

The molecules are further apart in ice than in water. ☐

The molecules are the same distance apart in ice and in water. ☐

The molecules are closer together in ice than in water. ☐

1 mark

Maximum 6 marks

Q1.

(a) It melts **or** is melting

*accept 'it starts to melt' or 'thaws' **or** 'turns to liquid' **or** 'changes state'*

*do **not** accept 'it is between solid and liquid'*

1 (L5)

(b) it is room temperature

accept no more energy being put in'

1 (L5)

(c) (i) They vibrate around fixed points **A**

*if more than one box is labelled **A**, award no mark*

1 (L6)

(ii) They move past each other and are close together **C**

*if more than one box is labelled **C**, award no mark*
disregard any ticks

1 (L6)

(d) (i) Ice is less dense than water. ✓

if more than one box is ticked, award no mark

1 (L6)

(ii) The molecules are further apart in ice than in water. ✓

if more than one box is ticked, award no mark

1 (L6)

[6]